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3. Design a CPU scheduling program with C using First Come First Served technique with the following considerations.

a. All processes are activated at time 0.

b. Assume that no process waits on I/O devices.

Code:

```
main.c
1 #include <stdio.h>
2 int main()
3 {
4     int n, i;
5     int bt[10], wt[10], tat[10];
6     float avg_wt = 0, avg_tat = 0;
7     printf("Enter number of processes: ");
8     scanf("%d", &n);
9     for(i = 0; i < n; i++)
10    {
11        printf("Enter Burst Time for P%d: ", i + 1);
12        scanf("%d", &bt[i]);
13    }
14    wt[0] = 0;
15    for(i = 1; i < n; i++)
16    {
17        wt[i] = wt[i - 1] + bt[i - 1];
18    }
19    for(i = 0; i < n; i++)
20    {
21        tat[i] = wt[i] + bt[i];
22        avg_wt += wt[i];
23        avg_tat += tat[i];
24    }
25    printf("\nProcess\tBT\tWT\tTAT\n");
26    for(i = 0; i < n; i++)
27    {
28        printf("P%d\t%d\t%d\t%d\n",
29            i + 1, bt[i], wt[i], tat[i]);
30    }
31    printf("\nAverage Waiting Time = %.2f",
32        avg_wt / n);
33    printf("\nAverage Turnaround Time = %.2f",
34        avg_tat / n);
35    return 0;
36 }
```

Output:

```
Enter number of processes: 3
Enter Burst Time for P1: 5
Enter Burst Time for P2: 3
Enter Burst Time for P3: 8

Process BT  WT  TAT
P1  5  0  5
P2  3  5  8
P3  8  8  16

Average Waiting Time = 4.33
Average Turnaround Time = 9.67

=== Code Execution Successful ===
```